MILITARY SPECIFICATION

TRANSISTOR, NPN, SILICON
TYPE 2N1493

1. SCOPE

1.1 Scope.- This document covers the detail requirements for silicon, NPN, transistor for use in VHF circuit applications and having the following particular characteristics at T_A=+25°±3°C. (See 3.2 herein.):

	P.G. (at: f = 70 Mc V_{CC} = 50 Vdc I_C = 25 mAdc)	h_{fe} (at: f = 70 Mc V_{CC} = 20 Vdc I_C = 15 mAdc)	h_{FE} (at: V_{CE} = 20 Vdc I_E = 10 mAdc)	C_{ob} (at: f= 0.1 to 1.0 Mc V_{CB} = 20 Vdc I_E = 0)	$r_b'C_c$ (at: V_{CC}= 20 Vdc I_C= 10 mAdc)
	db	---	---	uuf	uusec
Minimum	10	2.5	50	---	---
Maximum	---	---	200	5.0	100

1.2 Maximum ratings.-

P_C 1/	V_{CBO}	V_{CEX}	V_{EBO}	θ_{J-C}	T_J	T_{stg}
W	Vdc	Vdc	Vdc	°C/W	°C	°C
3.5	100	100	4.5	50	+200	-65 to + 200

1/ This power-dissipation rating is for 1,000 hours expected life at T_A= +25° ± 3°C. For power ratings at ambient temperatures above + 25°C, see Fig. 2 herein.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

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SPECIFICATIONS

MILITARY

MIL-S-19500 Semiconductor Devices, General Specification For

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be specified when requesting copies.)

3. REQUIREMENTS

3.1 General. - Requirements for the transistor shall be in accordance with Specification MIL-S-19500 and as otherwise specified herein.

3.2 Abbreviations and symbols used herein are defined in Specification MIL-S-19500 and as follows:

P_C ..... Collector power dissipation
P.G..... Large signal, high-frequency power gain
$r_b{}'C_c$..... High-frequency base resistance-collector capacitance product
V_{CEX}..... Collector-to-emitter voltage, emitter-to-base reverse biased

3.3 Design and construction. - The transistor shall be of the design, construction, and physical dimensions specified in Figure 1 herein.

3.3.1 Lead arrangement. - The lead arrangement on the transistor shall be as indicated in Figure 1 herein.

3.3.2 Operating position. The transistor shall be capable of proper operation in any position.

3.4 Performance characteristics. - The transistor performance characteristics shall be as specified in Tables I and II herein.

3.5 Marking. - The transistors shall be marked in accordance with Specification MIL-S-19500 and as follows. In instances where the diminutive size or lack of suitable surface area on the device would prevent a marking accomplishment readable by the unaided eye, 20/20 vision, at eight inches distance from the device, such marking may be omitted directly on the device. All required marking shall be placed on the unit package.

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3.5.1 Type-designation marking. - Type-designation marking of transistors procured on Department of Army contracts, and which have passed Government inspection and comply with all requirements of this specification, shall consist of: "USA-manufacturer's qualification code letters -transistor designation." The letters "JAN" or any abbreviation thereof shall not be used. If any specification waiver has been granted, the combination "USA-manufacturer's qualification code letters" shall not be used to complete the type-designation marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and Acceptance Inspection. - Qualification and Acceptance Inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. Groups A and B inspection shall consist of the examinations and tests specified in Tables I and II respectively, herein. Acceptance inspection shall include inspection of Preparation For Delivery (see 5.1 herein.)

4.2.1 Specified LTPD for subgroups. - The LTPD specified for a subgroup in Tables I and II herein shall apply for all of the tests, combined, in the subgroup.

4.2.1.1 Sample size for subgroups. - The sample size shall be selected by the manufacturer and shall be within the Max. Acc. No. limit specified herein for the particular subgroup LTPD.

4.2.2 Reinspection. - For customer verification inspection, lots may not be rejected unless the sample size for the particular subgroup being reinspected is not smaller than that required for the Max. Acc. No. specified herein relative to the subgroup LTPD.

4.2.3 Destructive tests. - The Group B, Subgroups 2, 3, 4, 5, 6, 7, and 8 tests are considered destructive. However, the tests of Subgroups 2, 3, 5, 6, 7, and 8 can be considered non-destructive if sufficient evidence to that effect is presented to the Government inspection authority. Acceptable evidence, for example, would be repeating of all Subgroups 2, 3, 5, 6, 7, and 8 tests, ten times, without significant device degradation. This test repetition procedure need be done only once at inception of Acceptance Inspection, provided no change in design or of production techniques has been effected.

4.2.4 Disposition of sample units. - Sample units that have been subjected to Group B, Subgroup 4 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroups 2, 3, 5, 6, 7, and 8 tests not determined to be destructive tests may be delivered on the contract or order provided that, after Group B inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

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Table I. Group A inspection.

Test Method per MIL-STD-750	Examination or Test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		5	4				
2071	Visual and mechanical examination	---			---	---	---	---
	Subgroup 2		5	3				
3036	Collector-base cutoff current	Bias Cond. D V_{CB} = 12 Vdc I_E = 0			I_{CBO}	---	1.0	uAdc
3076	Forward-current transfer ratio	V_{CE} = 20 Vdc I_E = 10 mAdc			h_{FE}	50	200	---
3030	Collector-emitter voltage (emitter-base reverse biased)	V_{EB} = 0.5 Vdc I_C = 100 uAdc			V_{CEX}	100	---	Vdc
3001	Collector-base breakdown voltage	Bias Cond. D I_C = 100 uAdc I_E = 0			BV_{CBO}	100	---	Vdc
3026	Emitter-base breakdown voltage	Bias Cond. D I_E = 100 uAdc I_C = 0			BV_{EBO}	4.5	---	Vdc
	Subgroup 3		5	3				
3236	Output capacitance	V_{CB} = 20 Vdc I_E = 0 f = 0.1 to 1.0 Mc			C_{ob}	---	5.0	uuf

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Table I. Group A inspection - (Cont'd).

Test Method per MIL-STD-750	Examination or Test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3 -(Cont'd)							
3255	Large signal, high-frequency power gain	f = 70 Mc V_{CC} = 50 V I_C = 25 mA P_o = 500 mW Test circuit per Fig. 3 herein.			P.G.	10	---	db
3306	Forward-current transfer ratio	f = 70 Mc V_{CC} = 20 Vdc I_C = 15 mAdc			h_{fe}	2.5	---	---
---	Base resistance-collector capacitance product	V_{CC} = 20 V I_C = 10 mA Test circuit per Fig. 4 herein			$r_b'C_c$	---	100	uu sec
	Subgroup 4		10	3				
1/	High-temperature operation:	T_A = + 125° ± 3°C						
3036	Collector-base cutoff current	Bias Cond. D V_{CB} = 12 Vdc I_E = 0			I_{CBO}	---	5.0	uAdc
1/	Low-temperature operation:	T_A = -55°C						
3076	Forward-current transfer ratio	V_{CE} = 20 Vdc I_E = 10 mAdc			h_{FE}	17	---	---

1/ Test measurement shall be made after thermal equilibrium has been reached at the temperature specified.

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Table II. Group B inspection.

Test Method per MIL-STD-750	Examination or Test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1		10	3				
2066	Physical dimensions	---			---	---	---	---
	Subgroup 2		10	4				
2026	Solderability	---			---	---	---	---
1051	Temperature cycling	Test Cond. C.			---	---	---	---
1056	Thermal shock (glass strain)	Test Cond. A.			---	---	---	---
1021	Moisture resistance	---			---	---	---	---
	End-point tests:							
3036	Collector-base cutoff current	Bias Cond. D V_{CB} = 12 Vdc I_E = 0			I_{CBO}	---	2.0	uAdc
3001	Collector-base breakdown voltage	Bias Cond. D I_C = 200 uAdc I_E = 0			BV_{CBO}	100	--	Vdc
3076	Forward-current transfer ratio	V_{CE} = 20 Vdc I_E = 10 mAdc			h_{FE}	40	---	---

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Table II. Group B inspection - (Cont'd)

Test Method per MIL-STD-750	Examination or Test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 3		10	4				
2016	Shock	Non-operating G = 500 5 blows of 1 msec ea. in ea. orientation X1, Y1, Y2, Z1 (total = 20 blows)			---	---	---	---
2006	Constant acceleration (centrifuge)	G = 10,000			---	---	---	---
2046	Vibration fatigue	Non-operating			--	---	---	---
2056	Vibration, variable frequency	G = 10 f = 100 to 1000 cps			---	---	---	---
	End-point tests:							
	Same as for Subgroup 2 above							
	Subgroup 4		10	3				
2036	Lead fatigue	Test Cond. E; 1/			---	---	---	---
	Subgroup 5		10	3				
1041	Salt atmosphere (corrosion)	---			---	---	---	---
	End-point tests:							
	Same as for Subgroup 2, above							

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Table II. Group B inspection - (Cont'd)

Test Method per MIL-STD-750	Examination or Test	Conditions	LTPD	Max. Acc. No.	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 6		10	3				
3131	Thermal resistance	V_{CE} = 10 Vdc I_C = 200 mAdc			θ_{J-C}	---	50	°C/W
	Subgroup 7		λ = 10	---				
1031	High-temperature life (non-operating)	T_{stg} = + 200°C			---	---	---	---
	End-point tests:							
	Same as for Subgroup 2, above							
	Subgroup 8		λ = 10	---				
1026	Steady state operation life	T_A = + 100°C at heat sink P_C = 2.0 W V_{CB} = 50 V			---	---	---	---
	End-point tests:							
	Same as for Subgroup 2, above							

1/ Rejects from prior electrical tests from the same lot may be used for this test.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery shall be in accordance with Specification MIL-S-19500.

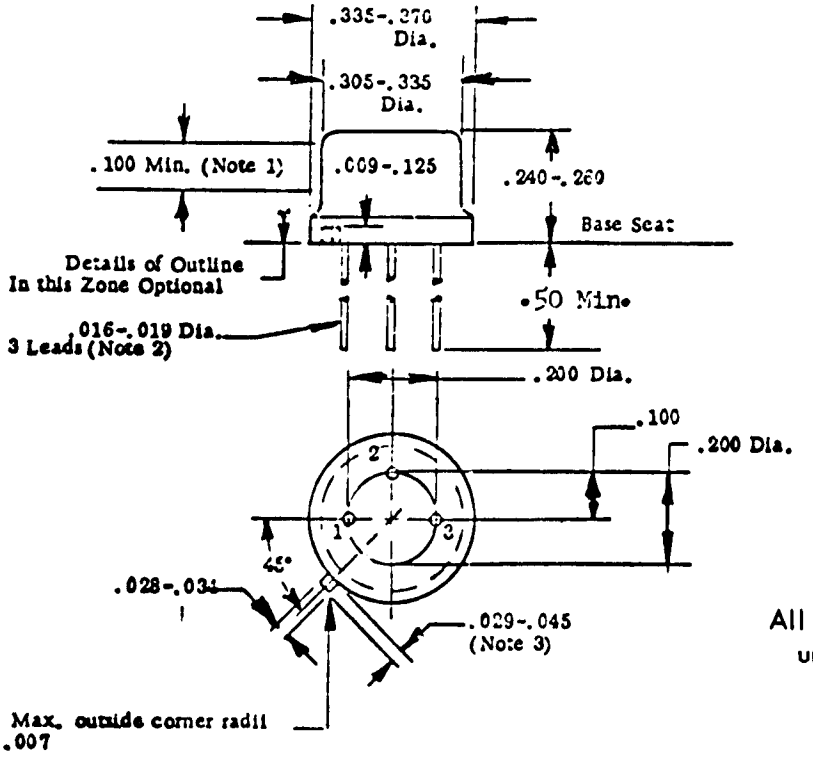
6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following exceptic ., are applicable to this specification.

6.2 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening for bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, supplement (Army), whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Chief, Standardization Engineering Division, U. S. Army Signal Materiel Support Agency, Fort Monmouth, New Jersey, Attention: SIGMS-PSM-3.

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LEADS:
1 Emitter
2 Base
3 Collector

All dimensions in inches unless otherwise specified

THIS DEVICE IS FOR SOCKETED, SINGLE-SIDED CIRCUIT-BOARD, WIRE-IN AND SIMILAR APPLICATIONS. WHERE USED IN DOUBLE-SIDED OR EYELETED CIRCUIT-BOARD OR SIMILAR APPLICATIONS WHERE SOLDER BRIDGING MAY OCCUR, A DIELECTRIC WASHER OR OTHER STANDOFF DEVICE MAY BE NECESSARY.

NOTE 1 - THIS ZONE IS CONTROLLED FOR AUTOMATIC HANDLING. THE VARIATION IN ACTUAL DIAMETER WITHIN THIS ZONE SHALL NOT EXCEED .010.

NOTE 2 - THE SPECIFIED LEAD DIAMETER APPLIES IN THE ZONE BETWEEN .050 AND .250 FROM THE BASE SEAT. BETWEEN .250 AND 1.5 A MAXIMUM OF .021 DIAMETER IS HELD. OUTSIDE OF THESE ZONES THE LEAD DIAMETER IS NOT CONTROLLED.

NOTE 3 - MEASURED FROM MAX. DIAMETER OF THE ACTUAL DEVICE.

NOTE 4 - LEADS HAVING MAXIMUM DIAMETER (.019) MEASURED IN GAGING PLANE .054 $^{+.001}_{-.000}$ BELOW THE SEATING PLANE OF THE DEVICE SHALL BE WITHIN .007 OF THEIR TRUE LOCATIONS RELATIVE TO MAXIMUM-WIDTH TAB.

NOTE 5 - THIS DEVICE MAY BE MEASURED BY DIRECT METHODS OR BY A GAGE AND GAGING PROCEDURE.

Figure 1. Outline and dimensions

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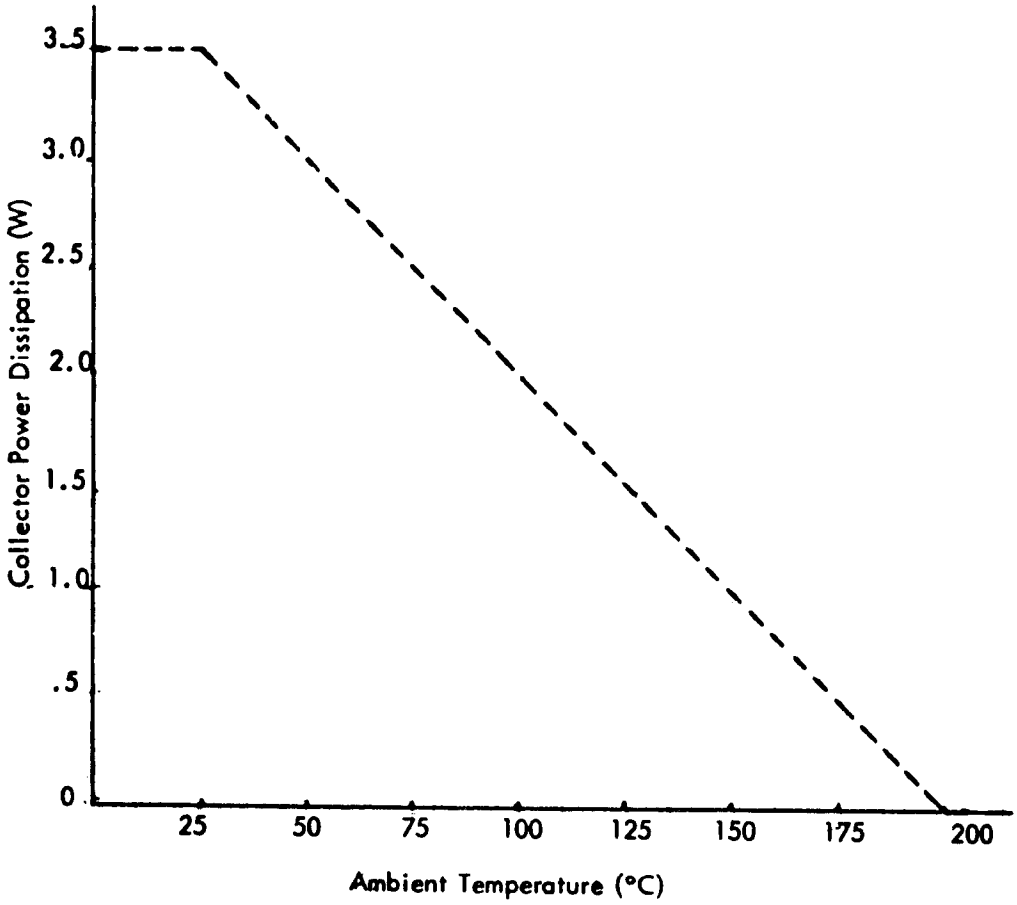
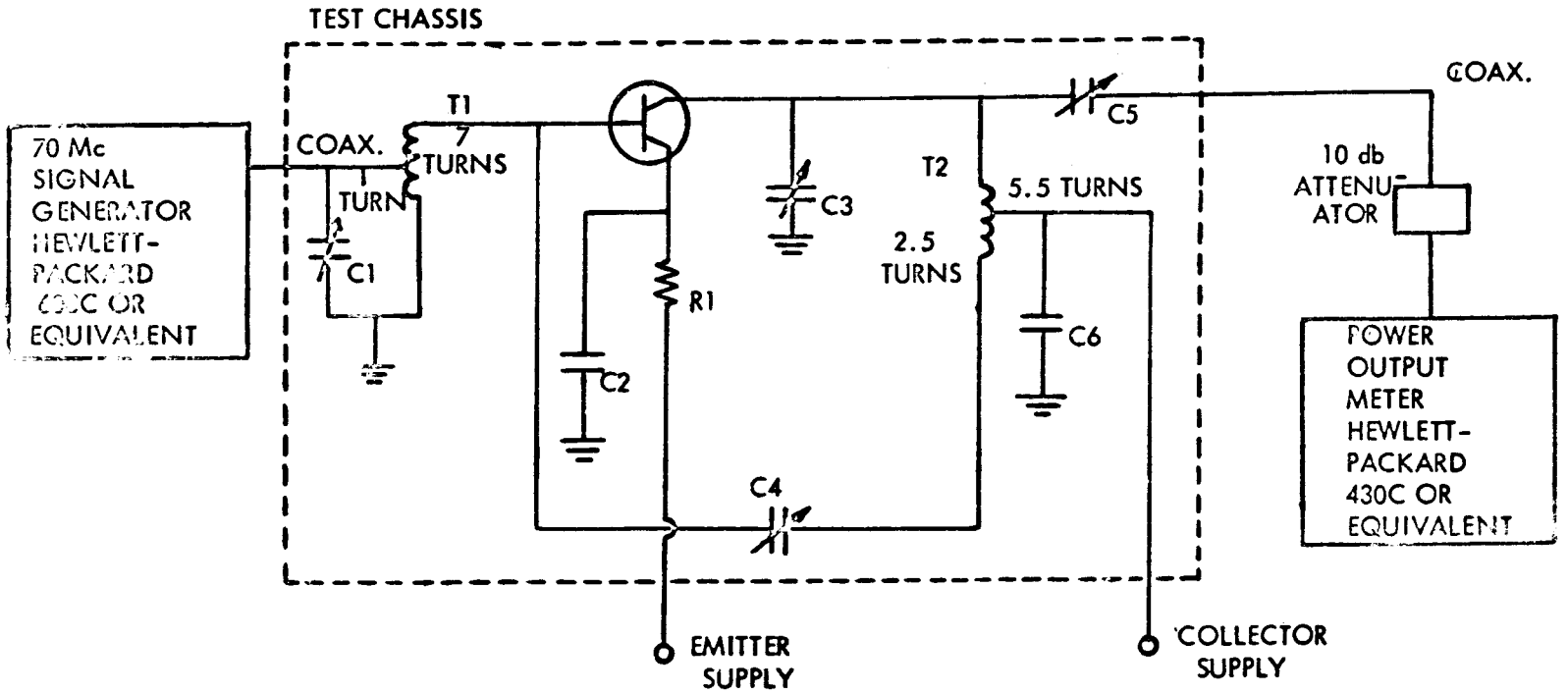


Figure 2. Power dissipation derating curve.

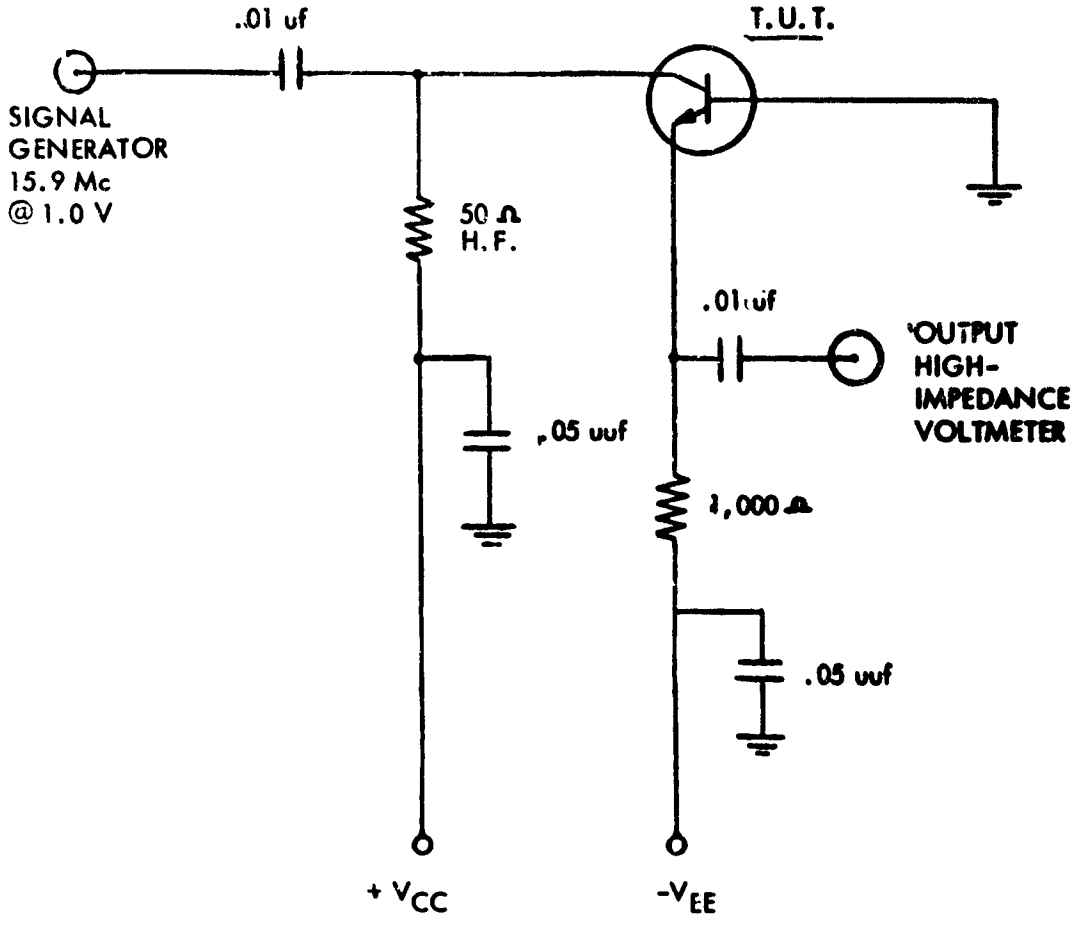
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C1 - Hammerlund HF-15 or equivalent.
C2 - .01 uf.
C3 - Hammerlund HF-15 or equivalent.
C4 - Arco 423 or equivalent.
C5 - Hammerlund HF-15 or equivalent.
C6 - .0025 uf.
R1 - 3KΩ-2 watt.
T1 - 8 turns #24 wire on a F303 toroid of Q3 material, General Ceramics Corp. (or equivalent) tapped at 1 turn.
T2 - 8 turns #24 wire on a CTC 3/8" ceramic coil form (or equivalent) tapped at 2.5 turns. (No slug used.)

Figure 3. Power Gain test circuit.

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$r_b{}'C_c$ (uu sec) = output (read directly on voltmeter) x 10^4

Figure 4. $r_b{}'C_c$ test circuit.

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